YACHAY TECH — SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Distributed Systems
Workshop 1: Client/Server

Semester II 2026
Aug 20, 2026
Prof. Francisco Hidrobo

Groups

This workshop will be held in groups of up to three students.

Workshop Activities

Using the example codes provided, for Server and Client (in python):

1. Test the programs on your own machine.
2. Test the programs on different hosts (Client and Server running on different hosts).
3. Test with two or more clients running concurrently. You should add multithreading handling in python for server and client (hint: use the multithreading module).
4. Change the Client code to send a certain amount of messages (the content of the messages can be randomly generated) to the server and terminate the communication. The number of messages should also be generated randomly.
5. Repeat steps 2 and 3, with the client version created in step 4.
6. Create C Language versions for Server and Client and repeat the tests.
7. Try the client and server in different languages.

Report

You should submit a report with code and results (screenshots). The workshop report should be clear and concise and include the following sections:

1. **Cover Page:** Include the workshop title, group members, course, and date.
2. **Objectives:** Briefly describe the main objectives of the workshop and what the group expects to demonstrate through the activities.
3. **Environment and Configuration:** Describe the hardware, operating systems, programming languages, software tools, and network configuration used to perform the tests.
4. **Implementation:** Briefly explain the Python and C client/server implementations and the modifications made for multithreading, multiple clients, and random message generation. Include only relevant code fragments when necessary.
5. **Experiments and Results:** Document each of the workshop activities separately. For each test, describe the configuration and procedure, report the result obtained, and provide screenshots as evidence.

6. **Results Summary:** Provide a table summarizing the tests performed, including the client and server languages, execution hosts, number of clients, and whether each test was successful.
7. **Problems and Solutions:** Describe any significant problems encountered during implementation or testing and explain how they were solved.
8. **Discussion:** Analyze the results and briefly discuss aspects such as concurrent clients, multithreading, differences between Python and C implementations, and communication between programs written in different languages.
9. **Conclusions:** Summarize the main results of the workshop and the knowledge gained from the implementation and experiments.
10. **References:** List the documentation, books, websites, or other resources consulted during the workshop.
11. **Appendix – Source Code:** Include the complete and properly organized source code for all Python and C client/server implementations. (Please use github)